

Lab 5 – Exploratory Assessment: Full-Stack Data Analytics Project

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Course: CPAN 212 – Modern Web Technologies

Lab 5: Exploratory Assessment

Chosen Focus: Data Analytics / Science

Coming from an Economics background, I've always been interested in patterns and insights hidden in data. I wanted to use this lab as a way to bridge what I've learned in CPAN 212 (frontend, backend, and databases) with my academic strengths, and explore Data Analytics as a potential career path.

What I Did

I downloaded a Netflix dataset and used Python (with Pandas) to perform an exploratory data analysis. I calculated summary statistics such as the number of total entries, content type distribution (Movies vs TV Shows), top producing countries, most common genres, and content distribution by release year. To store these results, I connected Python to MongoDB using the pymongo library.

Once the data was stored in MongoDB, I created a Node.js + Express backend to serve the results via a RESTful API. A frontend HTML/JavaScript page was built to fetch this data from the API and display the insights as a simple dashboard. This approach tied together Python for data analysis with the backend and frontend concepts we've been learning in this course.

Technologies Used

- **Python** (for analysis using Pandas and pymongo)
- **MongoDB Atlas** (to store and retrieve the results)
- **Node.js & Express** (for creating a RESTful API)
- **HTML/CSS/JavaScript** (to build the dashboard frontend)

Key Findings from the Dataset

- The dataset contained 8807 total entries, with a breakdown of 6131 Movies and 2676 TV Shows.
- The United States was the top content-producing country, followed by India, the United Kingdom, Japan, and South Korea.
- The most frequent genres included Dramas, International Movies, Documentaries, Stand-Up Comedy, and Independent Films.
- Netflix's content production ramped up significantly starting around 2015, peaking in the years 2016–2019.
- The release year distribution provided a clear visualization of Netflix's global content expansion in recent years.

What I Learned

- How to perform data cleaning and summarization using Python and Pandas
- How to use pymongo to insert structured data into MongoDB
- How to build an Express API to expose backend data to a frontend
- How to fetch and display data dynamically using frontend JavaScript
- How to bridge data science concepts with full-stack web development practices

Reflection

Although CPAN 212 focused primarily on JavaScript, Node.js, and MongoDB, I intentionally chose to use Python for the analysis portion of this project because it is the industry standard in Data Analytics, a field that strongly aligns with my background in Economics and my future career interests. I then integrated the output into a Node.js +

Express backend and connected it to MongoDB, which reflects the full-stack development principles we've been learning in this course.

By the end of this project, I felt a deeper connection between my academic interests and the practical tools we've used in class. It was fulfilling to see a real dataset transformed into a working dashboard and has motivated me to continue developing my skills in data analysis and backend integration.